

Subject Description Form

Subject Code	ITC6866
Subject Title	Textile Science V: Colour Science & Technology
Credit Value	3
Level	6
Pre-requisite / Co-requisite / Exclusion	Nil
Objectives	This subject is designed for the research students who are interested in the in-depth knowledge of colour science and state-of-the-art colour technology
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - Understanding of human colour vision in relation to colour perception - Have a sound knowledge on CIE colorimetry system - Describe and apply advanced colour difference formulae and its parametric effects in colour evaluation and colour control - Describe colour appearance and its usage in various media - Understand colour communication and colour management under multi-media environment, colour reproduction and its theoretical background
Subject Synopsis/ Indicative Syllabus	(I) Human colour vision Human colour perception and its interaction with light and objects of various media and surface conditions, phenomena related to the human colour perception, defective colour visions and its identification. (II) CIE colorimetry system CIE standard illuminants and observers, CIE colour specification system, CIE uniform colour space, Instrumental colour measurement and its optical arrangements. (III) Colour difference evaluation and colour appearance Visual and instrumental colour quality evaluation using respective methods such as grey scale and colour difference formulae, colour tolerance and its establishment, parametric effects on colour difference, colour appearance and its influence by various factors as well as its applications in colour evaluation and colour presentation (IV) Colour Communication under multi-media environment and colour reproduction Colour communication via various colour systems and under multi-media environment, colour reproduction of physical objects based on Kubelka-Munk theory, colour reproduction of self-luminous objects in modern display devices, such as flat panel display.

Teaching/Learning Methodology	Self-learning under the guidance will be adopted, literature related to the subject matters will be introduced, experiments are required to be conducted covering the various topics focused by the subject, data analysis and interpretation of the experimental results are important as well as the presentation of the experimental and learning outcomes.																																													
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="6">Intended subject learning outcomes to be assessed</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th><th></th></tr><tr><td>1. Exam</td><td>55%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td></td></tr><tr><td>2. Course work</td><td>45%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td></td></tr><tr><td>Total</td><td>100%</td><td colspan="6"></td></tr></table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Examination and course work which are based on design and conducting the experimental works can provide independent thinking and problem solving experience to students to understand and apply the subject matter.								Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed						a	b	c	d	e		1. Exam	55%	✓	✓	✓	✓	✓		2. Course work	45%	✓	✓	✓	✓	✓		Total	100%						
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Student Study Effort Expected	Class contact:																																													
	▪ Tutorial						18 Hrs.																																							
	▪ Student Presentation						2 Hrs.																																							
	Other student study effort:																																													
	▪ Self-learning						60 Hrs.																																							
	▪ Assignment and Laboratory						40 Hrs.																																							
	Total student study effort						120 Hrs.																																							
Reading List and References	<u>Books:</u> 1. J. H. Xin (Ed.), (2006). Total Colour Management in Textiles, Woodhead Publishing Ltd, Cambridge, England 2. Roy S Berns, (2000). Billmeyer and Saltzman's Principles of Color Technology, 3rd Edition, John Wiley & Sons. 3. Roderick McDonald, (1997). Colour Physics for Industry Society of Dyers & Colourists, 2nd Edition. 4. Hunt, R.W.G., (1991). Measuring Colour, 2nd Edition, Ellis Horwood Ltd. 5. Fairchild, M.D., (1998). Color Appearance Models, Addison Wesley Longman, Inc. 6. Hunt, R.W.G., (1995). The Reproduction of Colour, 5th Edition, Fountain Press, England. 7. Wyszecki, G. & Stiles, W.S., (1982). Colour Science, Concepts and Methods, Quantitative Data and Formulae, 2nd Edition, John Wiley & Sons. <u>Periodicals:</u> 1. Color Research and Applications, John Wiley, New York, USA. 2. Coloration Technology, the Society of Dyers and Colourists, United Kingdom.																																													